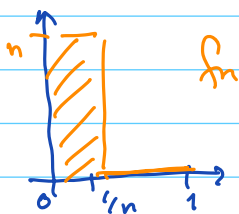


Homework 1 / 1 (cc) $f_n, f: [0,1] \rightarrow \mathbb{R}$ continuous

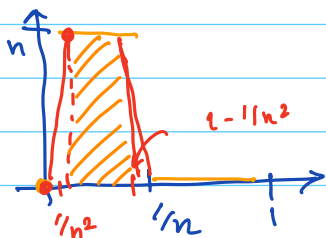
$f_n \rightarrow f$ pointwise, $\int_0^1 f_n \neq \int_0^1 f$



$$\int_0^1 f_n = 1$$

$$f_n(x) := \begin{cases} n & \text{if } 0 \leq x < 1/n \\ 0 & \text{if } x = 0 \\ 0 & \text{if } 1/n \leq x \leq 1 \end{cases}$$

$f_n \rightarrow f = 0$, $\int f_n = 1$, $\int f = 0$



$$f_n(x) := \begin{cases} \text{linear} & 0 \leq x < 1/n^2 \\ \text{linear} & 1/n^2 \leq x < 1/n \\ 0 & x \geq 1/n \end{cases}$$

————— // —————

Proposition 4.6 [Measure of an Increasing Function]

$(X, \mathcal{M}, \mu)$ measure space, $E_n \in \mathcal{M}$, $n \in \mathbb{N}$
 $E_n \subseteq E_{n+1}$, $n \in \mathbb{N}$ then

$$\left[\begin{array}{l} \subseteq \\ \cap \\ \neq \end{array} \right]$$

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

Proof If $\exists n \in \mathbb{N}$ s.t. $\mu(E_n) = +\infty$ then

monotonicity $\Rightarrow \forall m \geq n \quad \mu(E_m) \geq \mu(E_n) = +\infty$

$$\mu\left(\bigcup_{m=1}^{\infty} E_m\right) \geq \mu(E_n) = +\infty \quad \begin{array}{l} \text{done!} \\ \hline (\infty = \infty) \end{array}$$

Assume $\forall n \in \mathbb{N} \quad \mu(E_n) < +\infty$.

$$\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} (E_n \setminus E_{n-1}), \quad E_0 := \emptyset$$

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu\left(\bigcup_{n=1}^{\infty} (E_n \setminus E_{n-1})\right) = \sum_{n=1}^{\infty} \mu(E_n \setminus E_{n-1})$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \mu(E_k \setminus E_{k-1}) \quad \downarrow \text{ (4.3 (cc))}$$

$$= \lim_{k \rightarrow \infty} \sum_{n=1}^k [\mu(E_n) - \mu(E_{n-1})]$$

$$= \lim_{k \rightarrow \infty} \mu(E_k). \quad \blacksquare$$

Proposition 4.7 [Measure of a Decreasing Sequence]

$(X, \mathcal{M}, \mu)$ measure space, $E_n \in \mathcal{M}$, $E_n \supseteq E_{n+1} \forall n \in \mathbb{N}$, $\mu(E_1) < \infty$. Then

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

Proof De Morgan's law:

$$E_1 \setminus \bigcap_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} (E_1 \setminus E_n)$$

$$\emptyset = E_1 \setminus E_1 \subset E_1 \setminus E_2 \subset \dots \subset E_1 \setminus E_n \subset \dots$$

$$\text{Prop 4.6} \Rightarrow \mu(E_1 \setminus \bigcap_{n=1}^{\infty} E_n) = \mu\left(\bigcup_{n=1}^{\infty} (E_1 \setminus E_n)\right)$$

$$\cancel{\mu(E_1)} - \mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_1 \setminus E_n).$$

$$(4.3 \text{ (a)}) = \lim_{n \rightarrow \infty} [\cancel{\mu(E_1)} - \mu(E_n)]$$

$$\Rightarrow \mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n). \quad \blacksquare$$

Def 4.8 Given a measure space $(X, \mathcal{M}, \mu)$, the measure μ

is said complete if $\forall E \in \mathcal{M}$ with $\mu(E) = 0$ then

$$F \in \mathcal{M} \quad \forall F \subseteq E.$$

Thm 4.8* [Borel-Cantelli]

$(X, \mathcal{M}, \mu)$ measure space, $E_n \in \mathcal{M}$, $\sum_{n=1}^{\infty} \mu(E_n) < \infty$.

Then almost every $x \in X$ belongs to at most finitely many E_n 's,
i.e.

$$\mu(\{x \in X : x \in E_n \text{ for infinitely many } n\}) = 0.$$

Proof $\{x \in X : x \in E_n \text{ for infinitely many } n\}$

$$= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k =: \limsup_{n \rightarrow \infty} E_n =: E$$

Suppose that $\mu(E) = \alpha > 0$.

$$\text{So } 0 < \alpha = \mu(E) = \mu\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k\right)$$

$$(\text{monotonicity}) \leq \mu\left(\bigcup_{k=n}^{\infty} E_k\right) \quad \forall n \in \mathbb{N}.$$

$$(\text{subadditivity}) \leq \sum_{k=n}^{\infty} \mu(E_k).$$

Since $\sum_{n=1}^{\infty} \mu(E_n) < \infty$ choose $n > 1$

$$\sum_{k=n}^{\infty} \mu(E_k) < \alpha/2 \quad \text{--- } \checkmark \quad \blacksquare$$

n.b. Still ok for outer measures.

Theorem 4.9 [Carathéodory]

$X \neq \emptyset$ set, $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ outer measure.

Define (see (3.5))

$$\mathcal{M}^* := \{E \subseteq X : E \text{ } \mu^* \text{-measurable}\}$$

Then $\mathcal{M}^*$ is a σ -algebra, $\mu^*|_{\mathcal{M}^*} : \mathcal{M}^* \rightarrow [0, \infty]$ is a
complete measure.

(Recall 3.2 (iv)).

Proof Step 1 WTS that $\mathcal{M}^*$ is a σ -algebra (Def 3.4, (i)-(iii))

(i) $\emptyset \in \mathcal{M}^*$

$$\text{let } F \subseteq X \quad \mu^*(F) = \underbrace{\mu^*(F \cap \emptyset)}_{\mu^*(\emptyset)} + \underbrace{\mu^*(F \setminus \emptyset)}_{\mu^*(F)} \quad \text{OK!}$$

(ii) $E \in \mathcal{M}^*$

? $X \setminus E \in \mathcal{M}^*$

$$\begin{aligned} \text{let } F \subseteq X \quad \mu^*(F) &= \mu^*(F \cap E) + \mu^*(F \setminus E) \\ &= \mu^*(F \cap (X \setminus E)) + \mu^*(F \cap (X \setminus E)) \end{aligned}$$

$$\Rightarrow X \setminus E \in \mathcal{M}^*$$

(iii)' claim

$$E_1, E_2 \in \mathcal{M}^* \Rightarrow E_1 \cup E_2 \in \mathcal{M}^*$$

[In Step 4 we extend to countable unions, i.e., (iii) ... done!]

$$\text{Fix } F \subseteq X. \quad \mu^*(F) = \mu^*(F \cap E_1) + \mu^*(F \setminus E_1)$$

$$\mu^*(F \setminus E_1) = \mu^*((F \setminus E_1) \cap E_2) + \mu^*((F \setminus E_1) \setminus E_2)$$

Hence,

$$\mu^*(F) = \mu^*(F \cap E_1) + \mu^*((F \setminus E_1) \cap E_2) + \mu^*((F \setminus E_1) \setminus E_2)$$

$$(\text{Subadditivity}) \geq \mu^*((F \cap E_1) \cup ((F \setminus E_1) \cap E_2)) + \mu^*((F \setminus E_1) \setminus E_2)$$

$$= \mu^*(F \cap (E_1 \cup E_2)) + \mu^*(F \setminus (E_1 \cup E_2))$$

Step 2 $\mu^*|_{\mathcal{M}^*}$ is finite additive

$E_1, E_2 \in \mathcal{M}^*$ disjoint sets, let $F \subseteq X$.

$$\begin{aligned} \mu^*(F \cap (E_1 \cup E_2)) &= \mu^*(F \cap (E_1 \cup E_2) \cap E_1) + \mu^*(F \cap (E_1 \cup E_2) \setminus E_1) \\ (\text{used } E_1 \cap E_2 &= \emptyset) &= \mu^*(F \cap E_1) + \mu^*(F \cap E_2), \end{aligned}$$

with $F = X \Rightarrow \mu^*(E_1 \cup E_2) = \mu^*(E_1) + \mu^*(E_2)$

By induction, if $E_1, \dots, E_n \in \mathcal{M}^*$ pairwise disjoint

$$\mu^*\left(F \cap \bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n \mu^*(F \cap E_k) \quad \forall F \subseteq X \quad (4.1).$$

Step 3 μ^* on $\mathcal{M}^*$ is countably additive.

$E_n \in \mathcal{M}^*$, $n \in \mathbb{N}$, pairwise disjoint

claim $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}^*$ and $\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu^*(E_n)$ (4.2)

let $F \subseteq X$. know $\forall m \in \mathbb{N} \quad \bigcup_{n=1}^m E_n \in \mathcal{M}^*$ ((iv)' step 1)

$$\text{so } \mu^*(F) = \mu^*\left(F \cap \left(\bigcup_{n=1}^{\infty} E_n\right)\right) + \mu^*\left(F \setminus \left(\bigcup_{n=1}^{\infty} E_n\right)\right)$$

$$(\text{by (4.1)}) = \sum_{n=1}^m \mu^*(F \cap E_n) + \mu^*\left(F \setminus \left(\bigcup_{n=1}^m E_n\right)\right)$$

$$(\text{monotonicity}) \geq \sum_{n=1}^m \mu^*(F \cap E_n) + \mu^*\left(F \setminus \left(\bigcup_{n=1}^{\infty} E_n\right)\right)$$

let $m \rightarrow \infty$; get

$$\mu^*(F) \geq \sum_{n=1}^{\infty} \mu^*(F \cap E_n) + \mu^*\left(F \setminus \left(\bigcup_{n=1}^{\infty} E_n\right)\right) \quad (4.3)$$

$$(\text{subadditivity}) \geq \mu^*\left(\bigcup_{n=1}^{\infty} (F \cap E_n)\right) + \mu^*\left(F \setminus \left(\bigcup_{n=1}^{\infty} E_n\right)\right)$$

$$= \mu^*\left(F \cap \left(\bigcup_{n=1}^{\infty} E_n\right)\right) + \mu^*\left(F \setminus \left(\bigcup_{n=1}^{\infty} E_n\right)\right).$$

(=)

Thus $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}^*$, and with $F := \bigcup_{n=1}^{\infty} E_n$

$$(4.3) \Rightarrow \mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \geq \sum_{n=1}^{\infty} \mu^*(E_n) \bullet$$

$\leq$ always!

Step 4 To complete Step 1, need to show

(iii) if $E_n \in \mathcal{M}^*$, $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}^*$?

$$E'_n := E_n, \quad E'_n := E_n \setminus \left(\bigcup_{j=1}^{n-1} E_j \right) \in \mathcal{M}^*$$

pairwise disjoint

$$\text{By (4.2)} \quad \bigcup_{n=1}^{\infty} E'_n \in \mathcal{M}^* \iff \bigcup_{n=1}^{\infty} E_n \in \mathcal{M}^*$$

Step 5 [Completeness]

claim $\mu^*|_{\mathcal{M}^*} : \mathcal{M}^* \rightarrow [0, +\infty]$ is complete.

let $E \in \mathcal{M}^*$, $\mu^*(E) < \infty$ let $F \subseteq E$

? $F \in \mathcal{M}^*$?

$$\text{let } A \subseteq X. \text{ Then } \mu^*(A) = \underbrace{\mu^*(A \cap E)}_{\subseteq E} + \mu^*(A \setminus E) = \mu^*(A \cap E). \quad (4.3^*)$$

$$\begin{array}{c} \text{+ monotonicity} \\ \hline = 0 \end{array}$$

$$\begin{aligned} \mu^*(A \cap F) &= \underbrace{\mu^*(A \cap F \cap E)}_{\subseteq E} + \mu^*(A \cap F \setminus E) \\ &\stackrel{\text{+ monotonicity}}{\leq} \underbrace{\mu^*(A \cap E)}_{=0} + \mu^*(A \cap F \setminus E) \\ &= \mu^*(A \cap E) \quad \downarrow F \subseteq E \quad (4.3^{**}) \end{aligned}$$

$$(4.3^*) + (4.3^{**}) \quad \mu^*(A) = \mu^*(A \cap F) = \underbrace{\mu^*(A \cap F)}_{\subseteq F} + \mu^*(A \setminus F).$$

Hence $F \in \mathcal{M}^*$.

$$\begin{array}{c} \text{monotonicity } \mu^*(F) = 0 \\ \hline 0 \end{array}$$

Example 4.10

(i) Def 3.14... the class of all $(\mathcal{L}^n)^*$ -measurable sets is the σ -algebra of Lebesgue measurable sets, $\mathcal{L}(\mathbb{R}^n)$.

Carathéodory Thm $\Rightarrow (\mathcal{L}^n)^*|_{\mathcal{L}(\mathbb{R}^n)}$ is a complete measure, called the n -dimensional Lebesgue measure, denoted by $\mathcal{L}^n$.

Prop 3.13 all Borel sets of $\mathbb{R}^n$ are Lebesgue measurable.

In particular, balls are Lebesgue measurable.

Scaling + translation invariance $\forall x \in \mathbb{R}^n, r > 0$

$$\mathcal{L}^n(B(x, r)) = \mathcal{L}^n(B(0, r)) = r^n \underbrace{\mathcal{L}^n(B(0, 1))}_{\alpha_n}$$

$$(\text{Remark 2.5}) \quad \alpha_n = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)}.$$

(ii) $I \subseteq \mathbb{R}$ open interval, $f: I \rightarrow \mathbb{R}$ increasing function,

Carathéodory Thm $\Rightarrow \mu_f^*$ restricted to the σ -algebra of all μ_f^* -measurable subsets of I is a complete measure denoted by μ_f (Lebesgue-Stieltjes measure generated by f).

(iii) Carathéodory Thm $\Rightarrow H_0^s$ restricted to the σ -algebra

of all H_0^s -measurable subsets of $\mathbb{R}^n$ is a complete measure

denoted by H^s , called s -dimensional Hausdorff measure.